

Welcome !

Program at a glance

Tuesday (May 24)

Introduction and update on RepeatExplorer

- Principles and history of RepeatExplorer
(J. Macas)
 - Diagnostic features of repetitive elements – Part I
(P. Neumann)
- Coffee break*
- Diagnostic features of repetitive elements – Part II
(P. Neumann)
 - RepeatExplorer tools for genome annotation
(P. Novák)

Lunch

Practical training I

Basic Protocols 1-4: troubleshooting & advanced data analysis

Dinner at CITYgastro restaurant

Wednesday (May 25)

Short talks

- Nicola Schmidt
- Ludwig Mann
- Zirlane Portugal da Costa

Coffee break

- Pol Fernández Mató
- Yennifer Mata-Sucre
- Nusrat Sultana

Lunch

Practical training II

- using RE output for annotating genome assemblies
- REXdb and DANTE
- structure-based annotation of complete LTR-retrotransposons

Thursday (May 26)

Short talks

- Matej Lexa
- Monika Čechová
- Sophie Maiwald
- Vratislav Peška

Coffee break

- Camila do Nascimento Moreira
- Veit Herklotz
- Alice Krumpolcová

Lunch

Practical training III

- reconstructing phylogenetic relationships
- local installation and running RE tools from a command line
- topics proposed by the participants & individual consultations

Protocols and tutorials

- RepeatExplorer principles and example protocols published
- Corresponding video tutorials available from  YouTube



Global analysis of repetitive DNA from unassembled sequence reads using RepeatExplorer2

Petr Novák, Pavel Neumann and Jiří Macas

RepeatExplorer2 is a novel version of a computational pipeline that uses graph-based sequencing reads for characterization of repetitive DNA in eukaryotes. The identification in any genome by using relatively small quantities of short sequence reads. Galaxy platform, which provides a user-friendly web interface for script execution. Compared to the original version of the pipeline, RepeatExplorer2 provides automatic annotation and quantification of the identified repeats. RepeatExplorer2 workflow and provide procedures for its application to (i) de novo species, (ii) comparative repeat analysis in a set of species, (iii) development of s experiments and (iv) identification of centromeric repeats based on ChIP-seq data. RepeatExplorer2 is available at <https://repeatexplorer-eu1.scripps.edu/>

Introduction

Complex eukaryotic genomes, including those of higher plants, are characterized by various types of repetitive sequences. These sequence elements (retrotransposons and DNA transposons) that are distributed in arrays of tandemly repeated satellite DNA that constitute repetitive genome regions. Differential amplification and retention of repetitive sequences during genome evolution, leading to, for example, the 2,400 plant species. Repetitive DNA is present even in the small up to 15% of nuclear DNA, and its proportion increases in animal genomes, reaching values as high as 85%. In addition, repetitive sequences facilitate rapid genome restructuring in multiple ways.

Consequently, repetitive DNA attracted the interest of researchers in genome biology, which in turn prompted the development of new technologies. This process was greatly accelerated by the introduction of NGS technologies, which can generate large volumes of genomic data rapidly. Because the lack of such data had been the main obstacle in the investigations of repetitive DNA in non-model species, the demand for computational tools specifically tailored for repetitive DNA analysis has increased. Several principal approaches have been introduced: (i) identification of repetitive DNA based on structural features of the analyzed sequences (reviewed in refs. 1-3). The unique identification of repetitive DNA based on graph-based repeat clustering, constitutes the

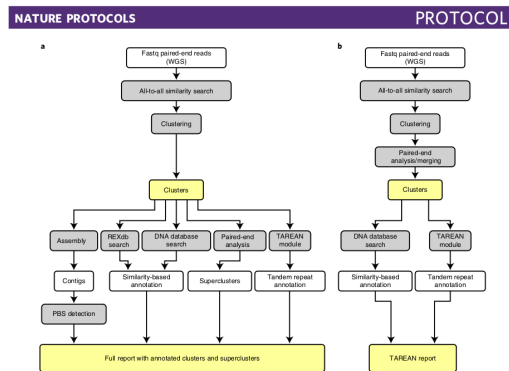


Fig. 1 | Schematic representation of RepeatExplorer (a) and TAREAN (b) pipelines. Analysis modules are represented by gray boxes, and input and output data are white, with the most important outputs highlighted in yellow. The RepeatExplorer pipeline is used in Procedures 1, 2 and 4; TAREAN is used in Procedure 3. WGS, whole-genome sequencing.

Overview of RepeatExplorer2

Recently, we released RepeatExplorer2, a new version of the pipeline that includes improvements of the existing programs and databases, as well as extended functionality due to inclusion of several novel tools. Although the basic workflow remains the same, a new module performs automatic annotation of the clusters based on the similarity hits to the reference databases and utilizing

NATURE PROTOCOLS

Box 6 | RepeatExplorer2 archive structure

Below is a description of the most important files within the output data, extracted from a zip archive. The extracted structure (folders are bold):

```
UnzippedGalaxyArchive
├── libdir
├── seqclust
├── small_clusters_assembly
├── custom_databases
├── reads
├── clustering
│   ├── clusters
│   ├── superclusters
│   └── hitsort.cls
├── logfile.txt
├── index.html
├── cluster_report.html
├── tarean_report.html
├── supercluster_report.html
├── summarized_annotation.html
├── PROFREP_CLASSIFICATION_TEMPLATE.csv
├── CLUSTER_TABLE.csv
├── SUPERCLUSTER_TABLE.csv
├── contigs.fasta
├── TAREAN_consensus_rank_1.fasta
├── TAREAN_consensus_rank_2.fasta
├── TAREAN_consensus_rank_3.fasta
├── TAREAN_consensus_rank_4.fasta
├── summary_histogram.png
├── documentation.html
├── HOW_TO_CITE.html
└── style1.css
```

which can be opened in a plain text editor to see how your analysis proceeded. The analysis went, as well as whether any error messages were printed out. The other HTML files are linked to index.html, provide more detailed information for a single cluster.

Basic information about clusters like the size of the cluster (i.e., the numbers and results of automatic annotation summarizing automatic annotation of superclusters.

Principles and history of RepeatExplorer

Principles and history of RepeatExplorer

2007

First paper
on repeat
clustering
from NGS
data

BMC Genomics



Research article

Open Access

**Repetitive DNA in the pea (*Pisum sativum* L.) genome:
comprehensive characterization using 454 sequencing and
comparison to soybean and *Medicago truncatula***

Jiří Macas*, Pavel Neumann and Alice Navrátilová

Address: Biology Centre ASCR, Institute of Plant Molecular Biology, Branišovská 31, Žeské Budějovice, CZ-37005, Czech Republic

Email: Jiří Macas* - macas@umbr.cas.cz; Pavel Neumann - neumann@umbr.cas.cz; Alice Navrátilová - navratil@umbr.cas.cz

* Corresponding author

Published: 21 November 2007

BMC Genomics 2007, 8:427 doi:10.1186/1471-2164-8-427

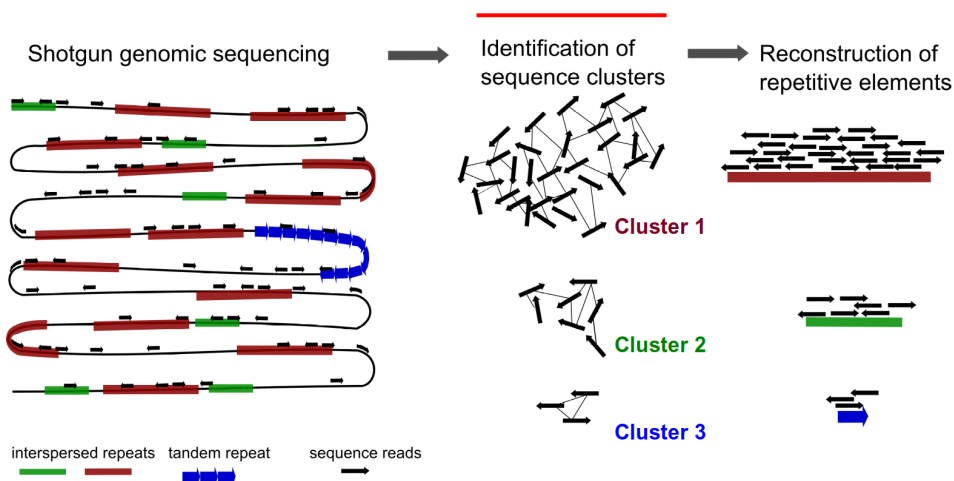
Received: 13 August 2007

Accepted: 21 November 2007

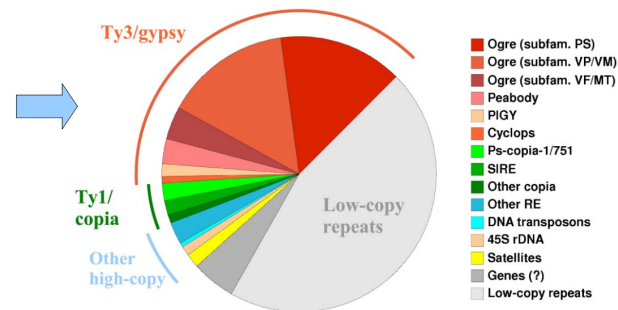
Principles and history of RepeatExplorer

2007

First paper
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clustering
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Proportions of various repeat types in a genome

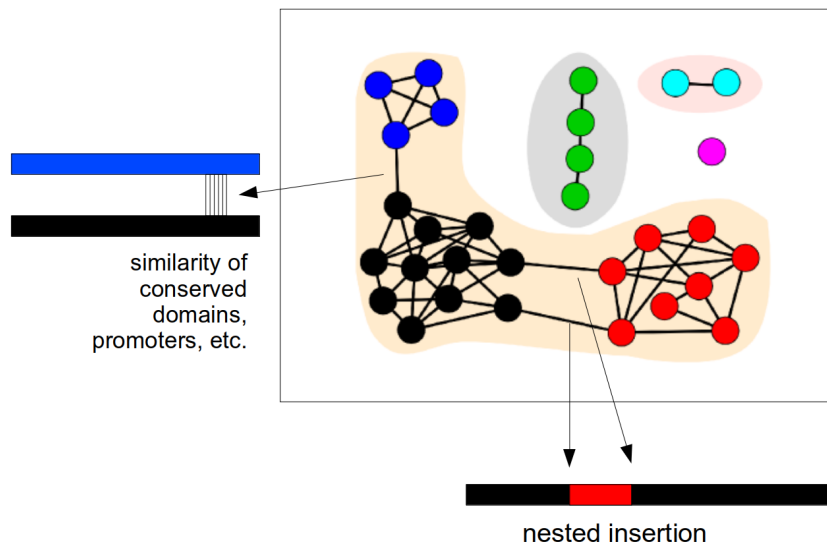


CLUSTER = a set of frequently overlapping reads = REPEAT FAMILY

Principles and history of RepeatExplorer

2007

First paper on repeat clustering from NGS data



Single linkage clustering => connected components

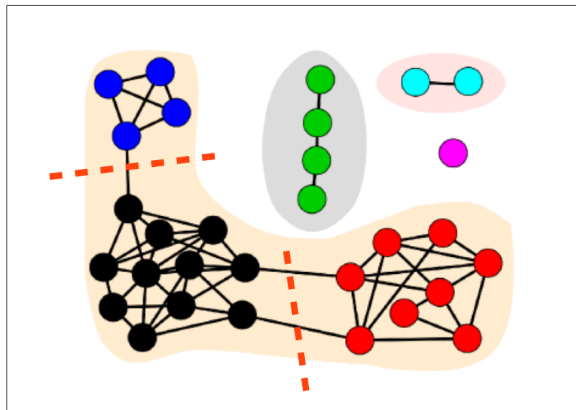
TGICL
(TIGR Gene Indices clustering tool)
Pertea et al., 2003

Principles and history of RepeatExplorer

2007 ... 2010

First paper
on repeat
clustering
from NGS
data

Introduction
of **graph-
based
clustering**
(Novak et
al. 2010)



Graph-based clustering

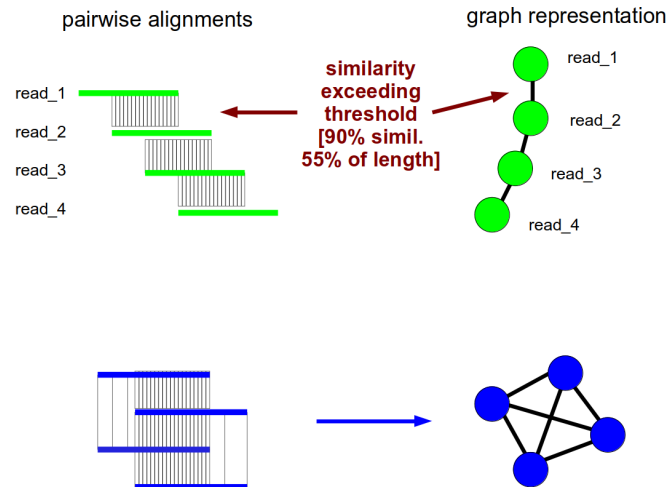
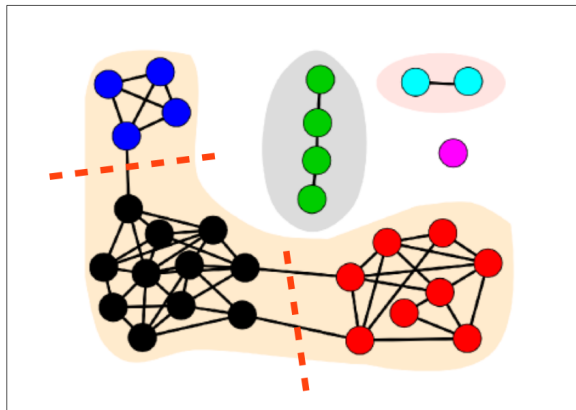
- Sequence overlaps between the reads are transformed to a graph where the **reads** are represented as **nodes** and their **similarities** as **edges** connecting the nodes
- Graph structure is examined to detect **communities of frequently connected nodes** which **are split to separate clusters**

Principles and history of RepeatExplorer

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Graph-based clustering

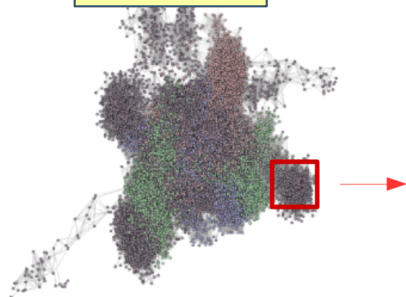
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Principles and history of RepeatExplorer

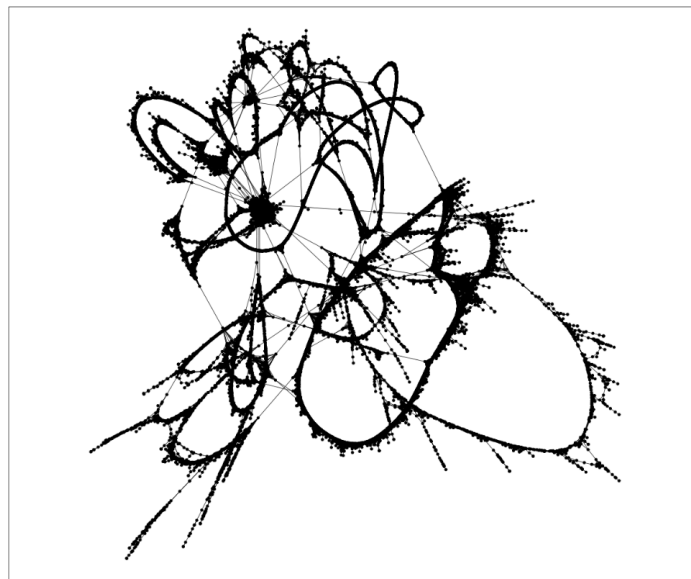
2007 ... 2010

First paper on repeat clustering from NGS data

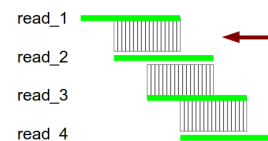
Introduction of **graph-based clustering** (Novak et al. 2010)



Virtual graphs used to analyze real data contain **up to millions of nodes** (reads)

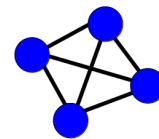
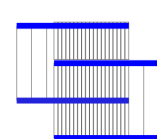
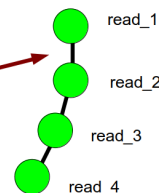


pairwise alignments



similarity exceeding threshold
[90% simil.
55% of length]

graph representation

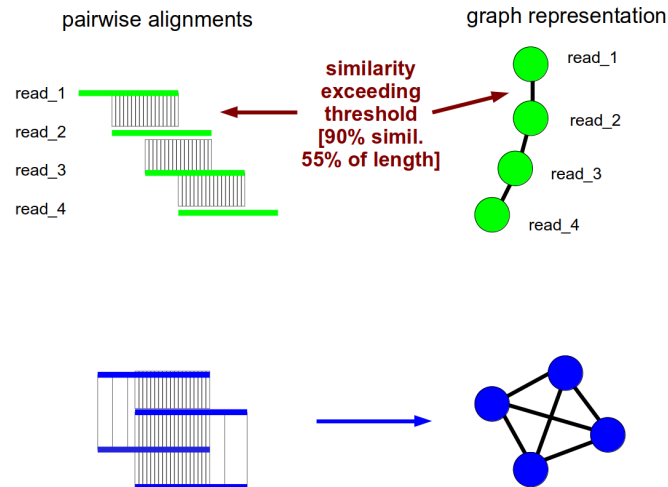
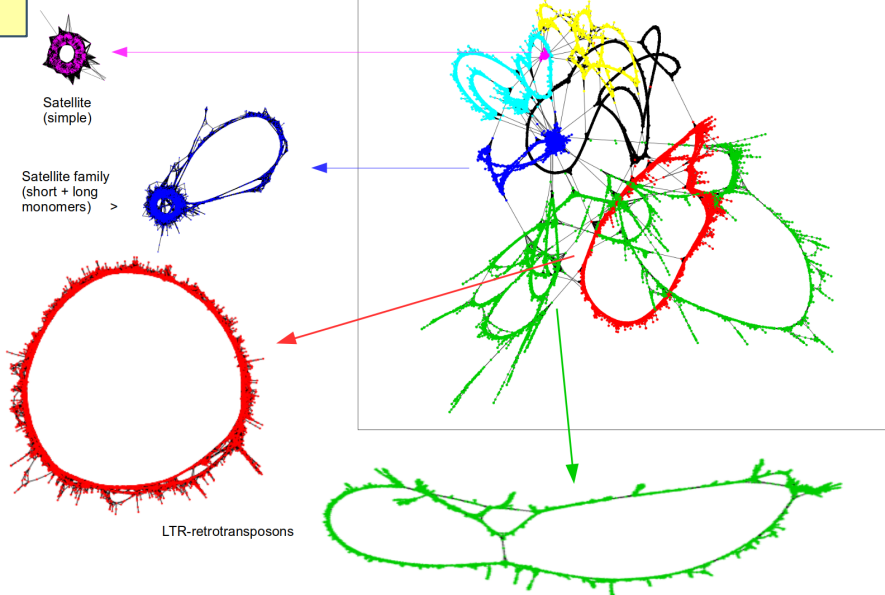


Principles and history of RepeatExplorer

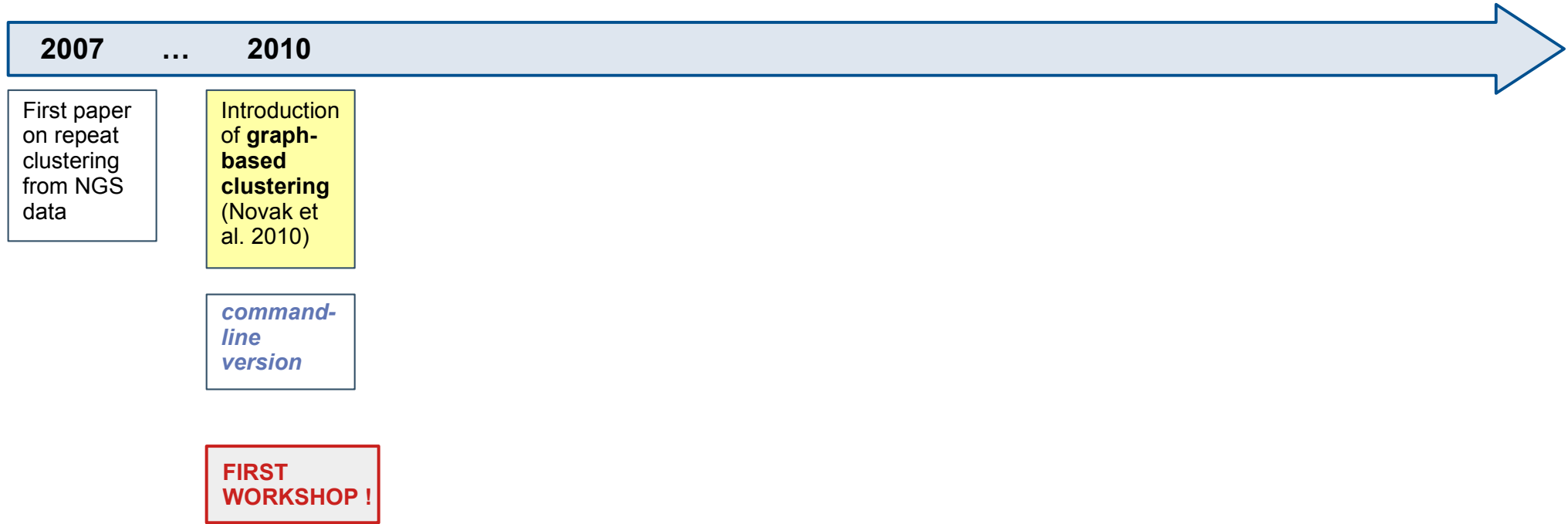
2007 ... 2010

First paper
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Introduction
of **graph-
based
clustering**
(Novak et
al. 2010)



Principles and history of RepeatExplorer



Principles and history of RepeatExplorer

2007

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2010

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2013

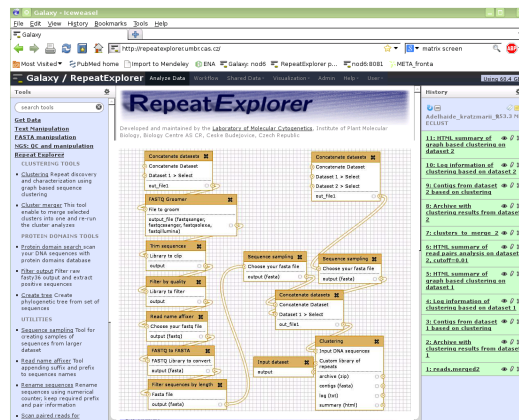
First paper on repeat clustering from NGS data

Introduction of **graph-based clustering** (Novak et al. 2010)

Repeat Explorer in Galaxy (Novak et al. 2013)

command-line version

Public web-based server



Principles and history of RepeatExplorer



Principles and history of RepeatExplorer

2007 ... 2010 ... 2013 2014 ... 2016 2017

First paper on repeat clustering from NGS data

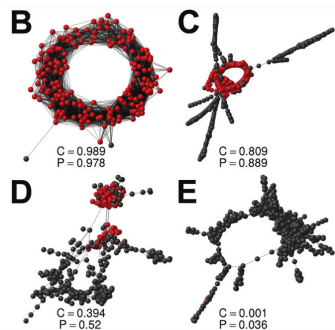
Introduction of **graph-based clustering** (Novak et al. 2010)

Repeat Explorer in Galaxy (Novak et al. 2013)



ELIXIR \$\$\$

TAREAN

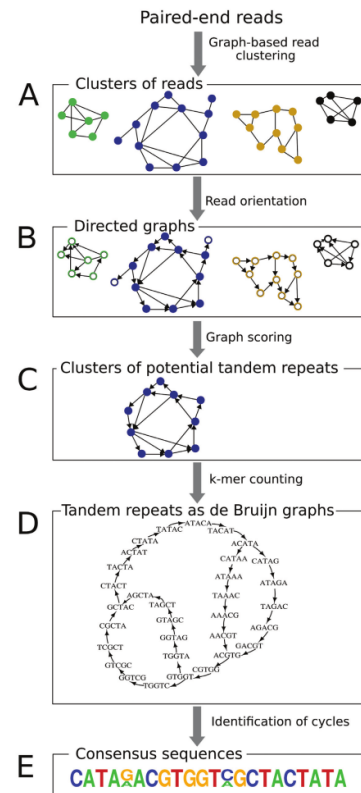


TAREAN: a computational tool for identification and characterization of satellite DNA from unassembled short reads

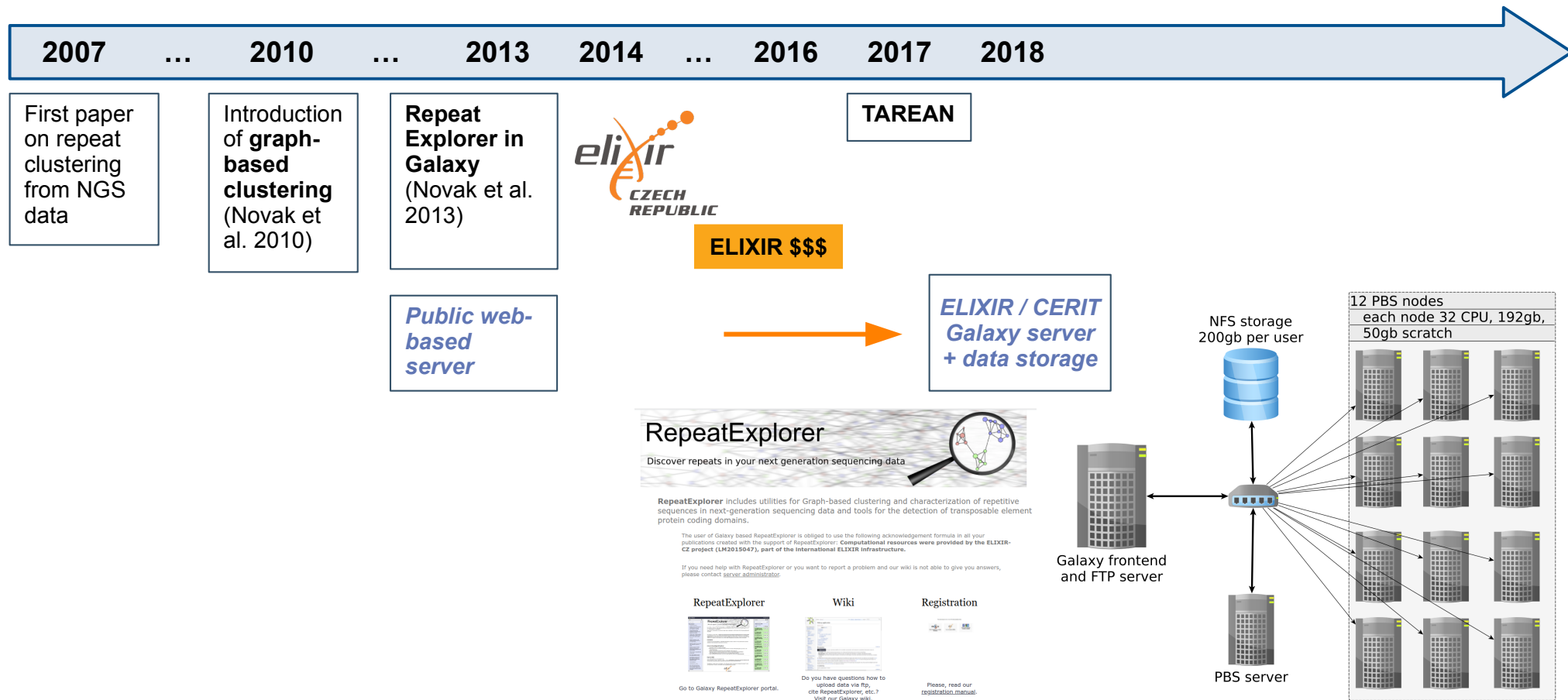
Petr Novák, Laura Ávila Robledillo, Andrea Koblížková, Iva Vrbová, Pavel Neumann and Jiří Macas*

Institute of Plant Molecular Biology, Biology Centre CAS, České Budějovice CZ-37005, Czech Republic

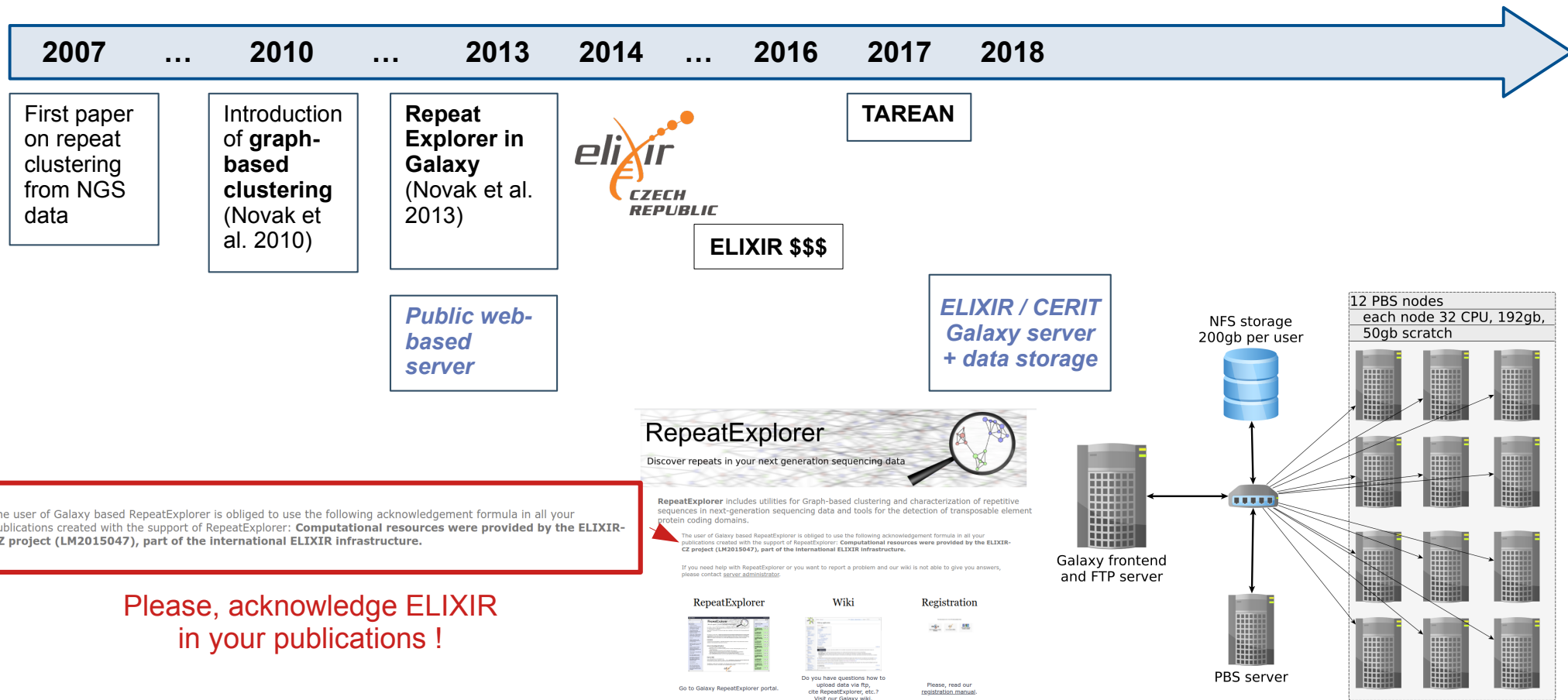
Nucleic Acids Research, 2017 **1**
doi: 10.1093/nar/gkx257



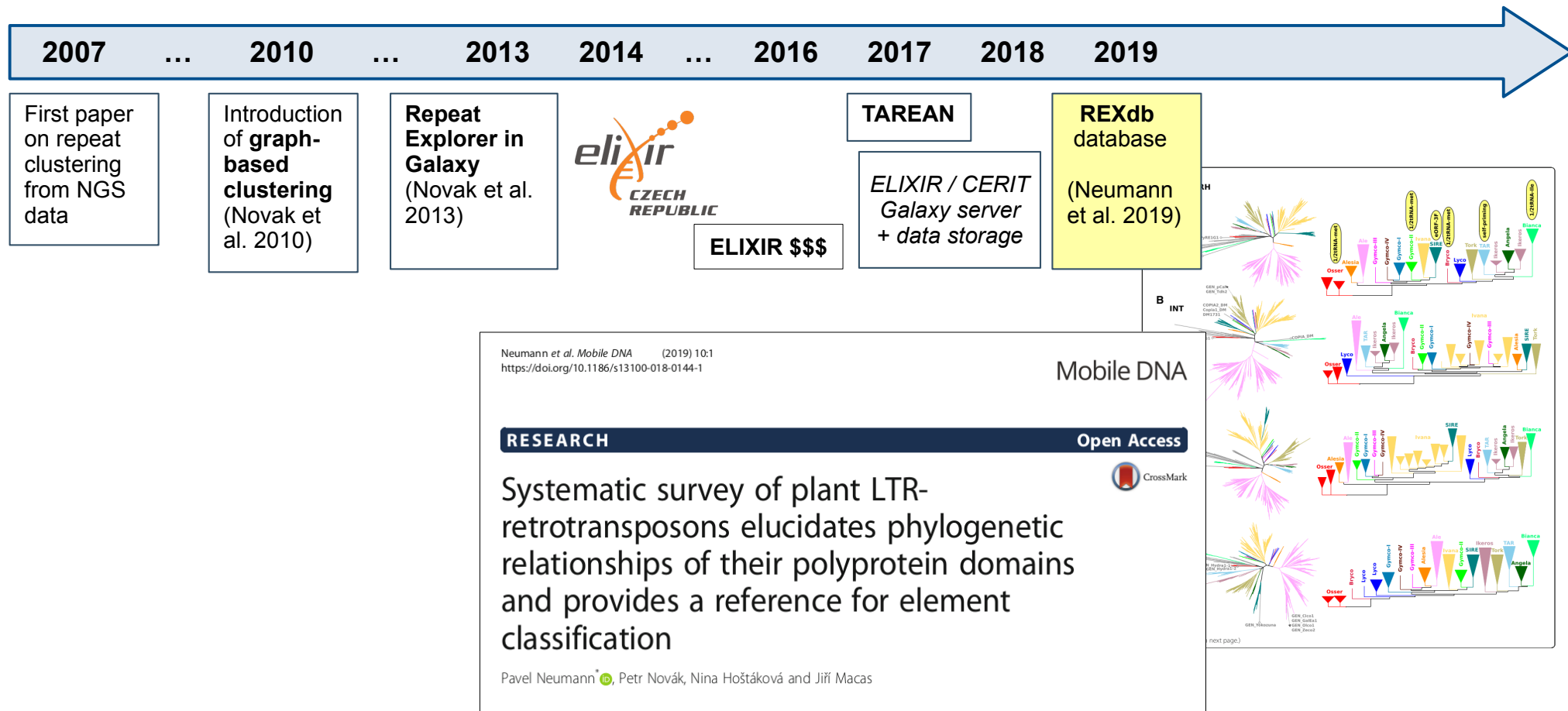
Principles and history of RepeatExplorer



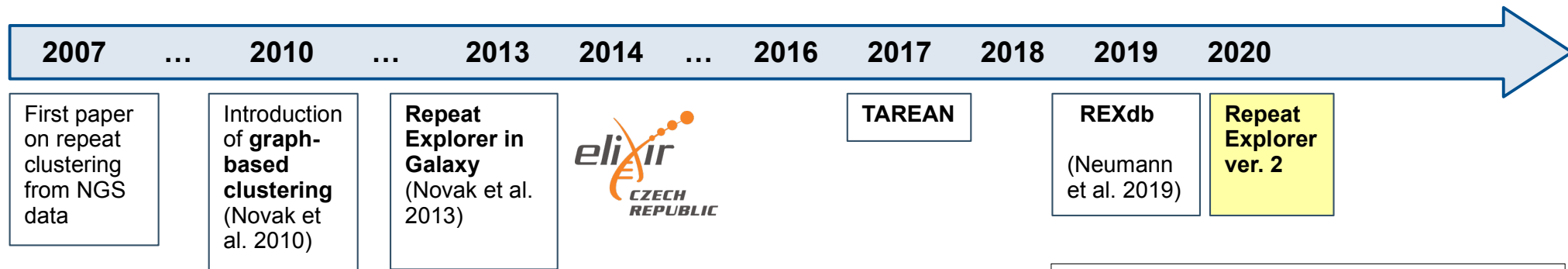
Principles and history of RepeatExplorer



Principles and history of RepeatExplorer



Principles and history of RepeatExplorer



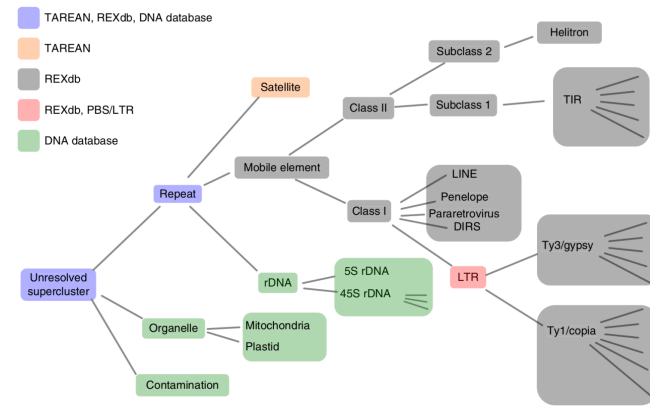
Neumann et al. *Mobile DNA* (2019) 10:1
<https://doi.org/10.1186/s13100-018-0144-1>

RESEARCH

Systematic survey of plant LTR-retrotransposons elucidates phylogenetic relationships of their polyprotein domain and provides a reference for element classification

Pavel Neumann*, Petr Novák, Nina Hošťáková and Jiří Macas

Decision tree for automatic annotation



Principles and history of RepeatExplorer

2007

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2010

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2013

2014

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2016

2017

2018

2019

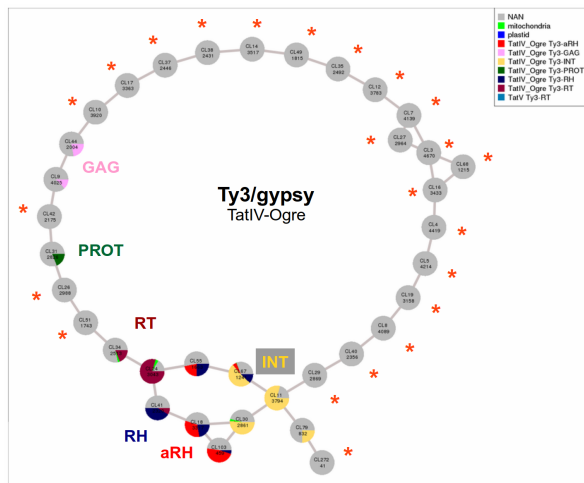
2020

TAREAN

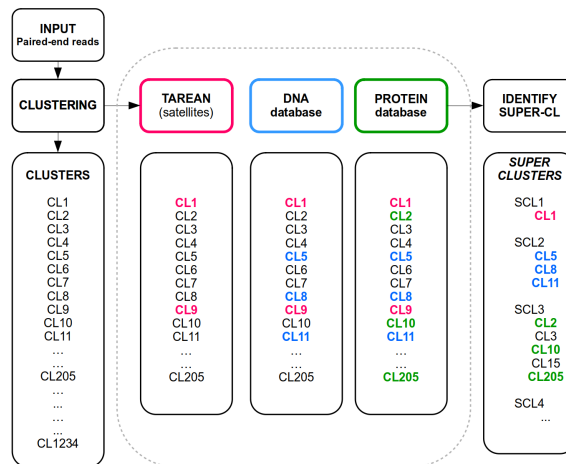
REXdb
(Neumann
et al. 2019)

Repeat
Explorer
ver. 2

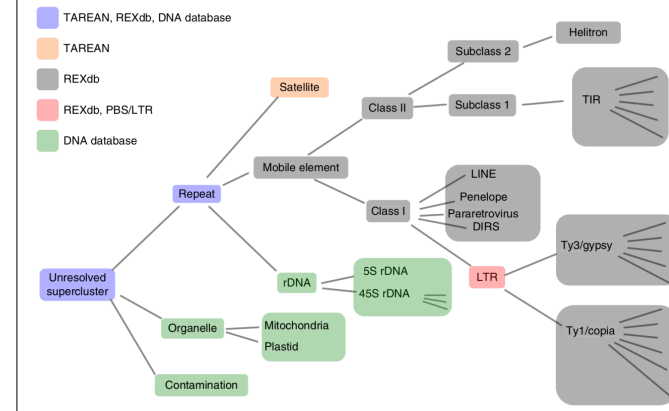
Superclusters provide more complete annotation



RepeatExplorer 2 – automatic detection of superclusters



Decision tree for automatic annotation



Principles and history of RepeatExplorer

2007

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2014

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2016

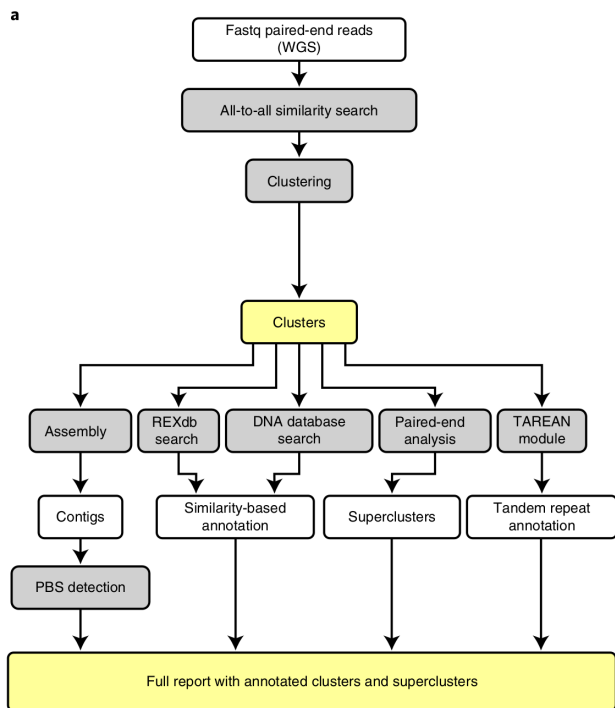
2017

2018

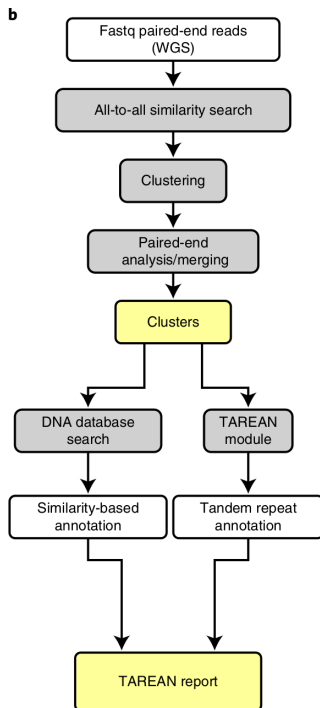
2019

2020

RepeatExplorer pipeline



TAREAN pipeline



TAREAN

REXdb
(Neumann
et al. 2019)

Repeat
Explorer
ver. 2

Principles and history of RepeatExplorer

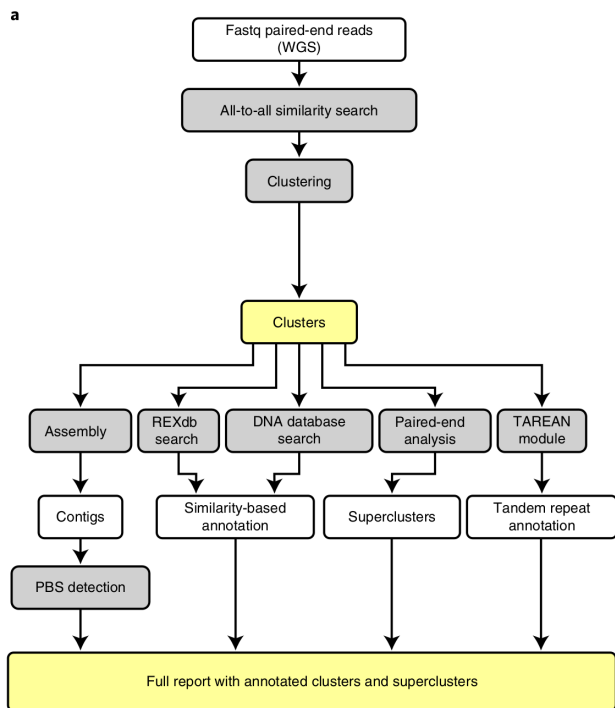
2007 ... 2010 ... 2013 2014 ... 2016 2017 2018 2019 2020 2021 2022

TAREAN

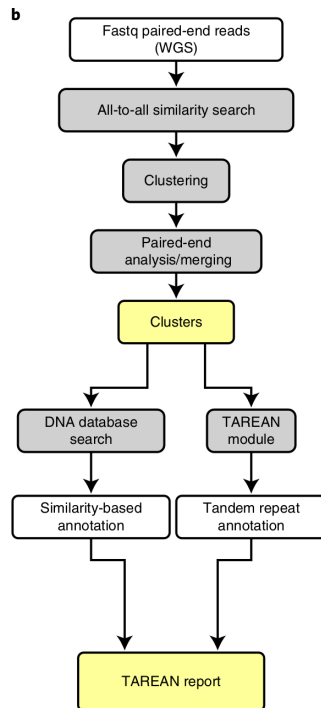
REXdb
(Neumann
et al. 2019)

Repeat
Explorer
ver. 2

RepeatExplorer pipeline



TAREAN pipeline



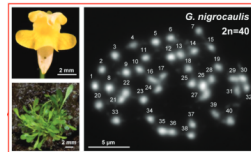
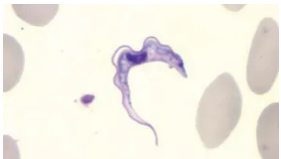
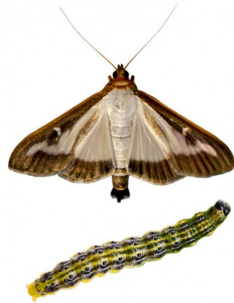
Additional tools:

- ChIP-seq Mapper
 - DANTE
 - Long reads
- Assembly annotation

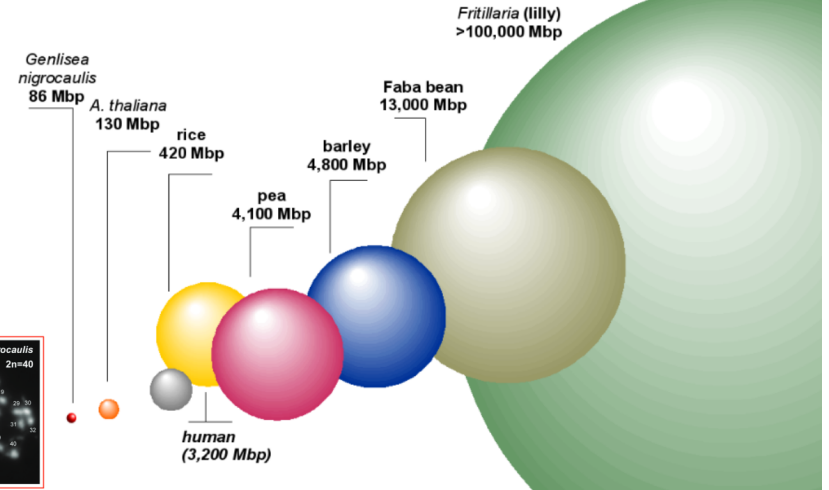
Applications

Applications

- Repeat composition
 - single species

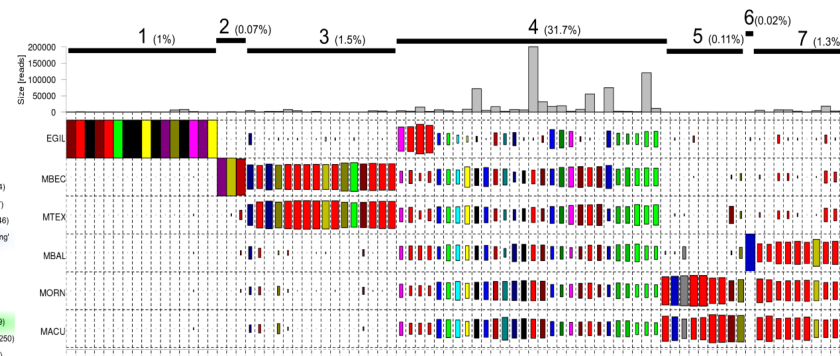
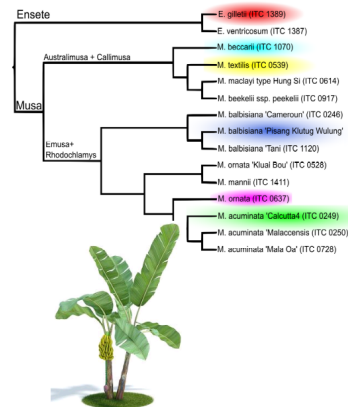


Plant species varying ~2,000-fold
in their genome sizes



Applications

- Repeat composition
 - single species
 - comparative analysis



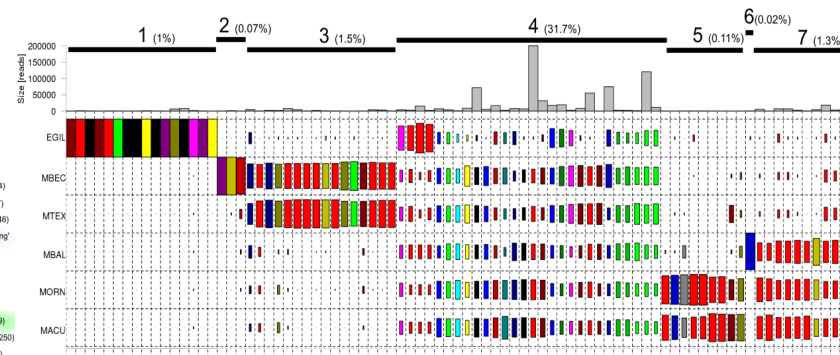
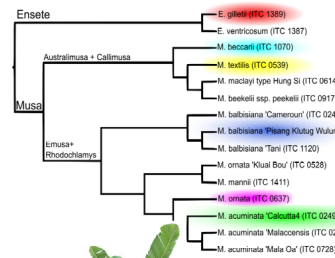
Genomic repeat abundances
contain phylogenetic signal

Dodsworth et al. (2015)
Syst. Biol. 64(1): 112-126

■ Maximus ■ Tat
 ■ Angela ■ Chromovirus/unspecified
 ■ Tork ■ TRIM
 ■ Ale ■ pararetrovirus
 ■ Ivana ■ LINE
 ■ TAR ■ DNA transposon
 ■ Reina ■ rDNA
 ■ Tekay ■ TR
 ■ Galadriel ■ unknown
 ■ CRM (Novak et al., 2014)

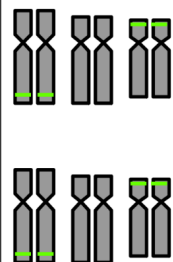
Applications

- Repeat composition
 - single species
 - comparative analysis



Identification of chromosome B-specific repeats

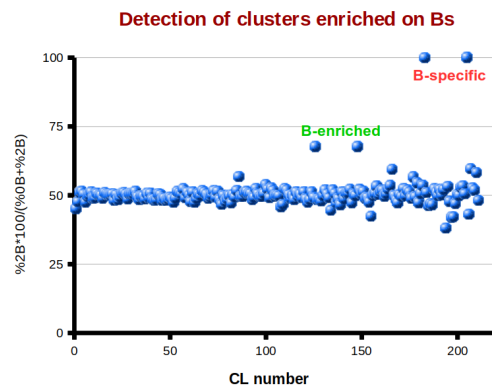
Comparative analysis of B+/- plants



Next-gen sequencing
As + Bs

Next-gen sequencing
As - 0 -

Mix coded reads and perform CLUSTERING



Genomic repeat abundances contain phylogenetic signal

Dodsworth et al. (2015)
Syst. Biol. 64(1): 112-126

■ Maximus ■ Tat
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 ■ Tork ■ TRIM
 ■ Ale ■ pararetrovirus
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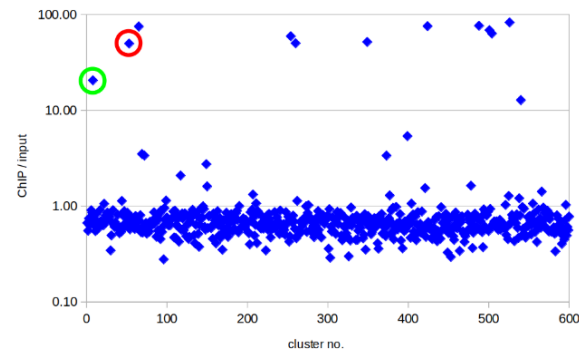
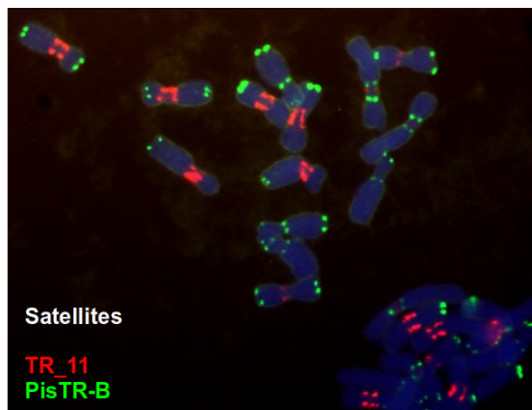
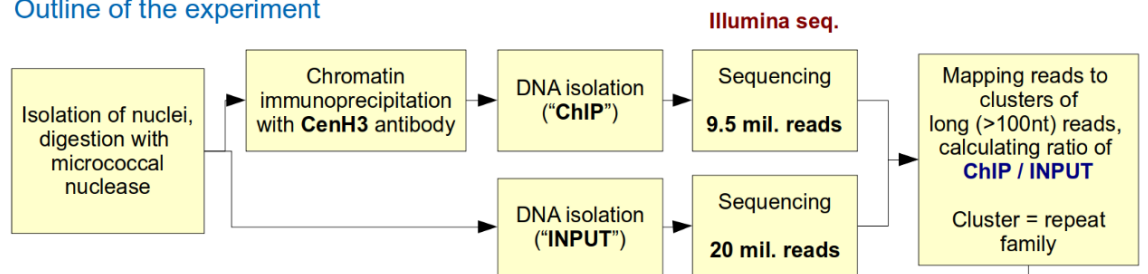
Applications

- Repeat composition
 - single species
 - comparative analysis

- Repeat clusters as a reference
 - ChIP-seq

Identification of centromeric repeats by ChIP-seq

Outline of the experiment



Neumann et al. (2012) *PLoS Genetics* 8:e1002777

Applications

Repeat composition

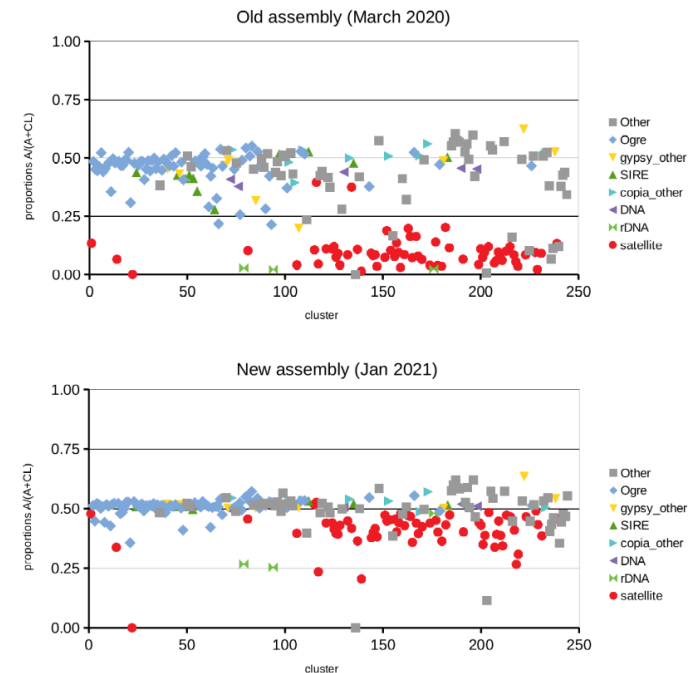
- single species
- comparative analysis

Repeat clusters as a reference

- ChIP-seq
- assembly

Assessing completeness of genome assemblies

				sum of hits		blasted fragments		CL/(CL+BLAST)	
				133232913		207900483			
Size	adjusted	Proportion [%]	Final_annotation	CL	hits				
85804	4.187		All/repeat/satellite/FabTR_FokI	1	6171084	2.968		0.13	
38954	1.901		All/repeat/mobile_element/Class	2	1343009	0.646		0.49	
37146	1.813		All/repeat/mobile_element/Class	3	3723933	1.791		0.45	
37051	1.808		All/repeat/mobile_element/Class	4	3091188	1.487		0.45	
36830	1.797		All/repeat/mobile_element/Class	5	3014725	1.450		0.46	
33547	1.637		All/repeat/mobile_element/Class	6	3221042	1.549		0.52	
32220	1.572		All/repeat/mobile_element/Class	7	3719091	1.789		0.44	
31149	1.520		All/repeat/mobile_element/Class	8	2574963	1.239		0.45	
30972	1.511		All/repeat/mobile_element/Class	9	2619726	1.260		0.47	
30361	1.482		All/repeat/mobile_element/Class	10	2791506	1.343		0.48	
29548	1.442		All/repeat/mobile_element/Class	11	2857979	1.375		0.35	
29227	1.426		All/repeat/mobile_element/Class	12	1649483	0.793		0.50	
27341	1.334		All/repeat/mobile_element/Class	13	2924386	1.407		0.48	
26964	1.316		All/repeat/mobile_element/Class	14	2544864	1.224		0.04	
26082	1.273		All/repeat/mobile_element/Class	15	189991	0.091		0.49	
25651	1.252		All/repeat/mobile_element/Class	16	2496449	1.201		0.97	
25147	1.227		All/repeat/mobile_element/Class	17	2280990	1.097		0.47	
23553	1.149		All/repeat/mobile_element/Class	18	2239097	1.077		0.49	
22914	1.118		All/repeat/mobile_element/Class	19	2304645	1.109		0.49	
22829	1.114		All/repeat/mobile_element/Class	20	2277830	1.072		0.52	
22696	1.108		All/repeat/mobile_element/Class	21	2547541	1.225		0.31	
22481	1.097		All/repeat/mobile_element/Class	22	1024482	0.493		0.00	
21364	1.043		All/repeat/mobile_element/Class	23	6	0.000		0.48	
20442	0.998		All/repeat/mobile_element/Class	24	1971842	0.948		0.44	
19605	0.957		All/repeat/mobile_element/Class	25	1610084	0.774		0.47	
19289	0.941		All/repeat/mobile_element/Class	26	1753755	0.844		0.48	
19234	0.939		All/repeat/mobile_element/Class	27	1835955	0.883		0.48	
18932	0.924		All/repeat/mobile_element/Class	28	1842540	0.886		0.41	
18345	0.895		All/repeat/mobile_element/Class	29	1318285	0.634		0.45	
18112	0.884		All/repeat/mobile_element/Class	30	1533938	0.738		0.49	
17991	0.878		All/repeat/mobile_element/Class	31	1753161	0.843		0.44	
17167	0.836		All/repeat/mobile_element/Class	32	1457392	0.701		0.49	
16736	0.816		All/repeat/mobile_element/Class	33	1663803	0.800		0.49	
			All/repeat/mobile_element/Class	34	1693468	0.815		0.49	



Applications

- Repeat composition
 - single species
 - comparative analysis
- Repeat clusters as a reference
 - ChIP-seq
 - assembly
 - reference databases for repeat annotation

Applications

Repeat composition

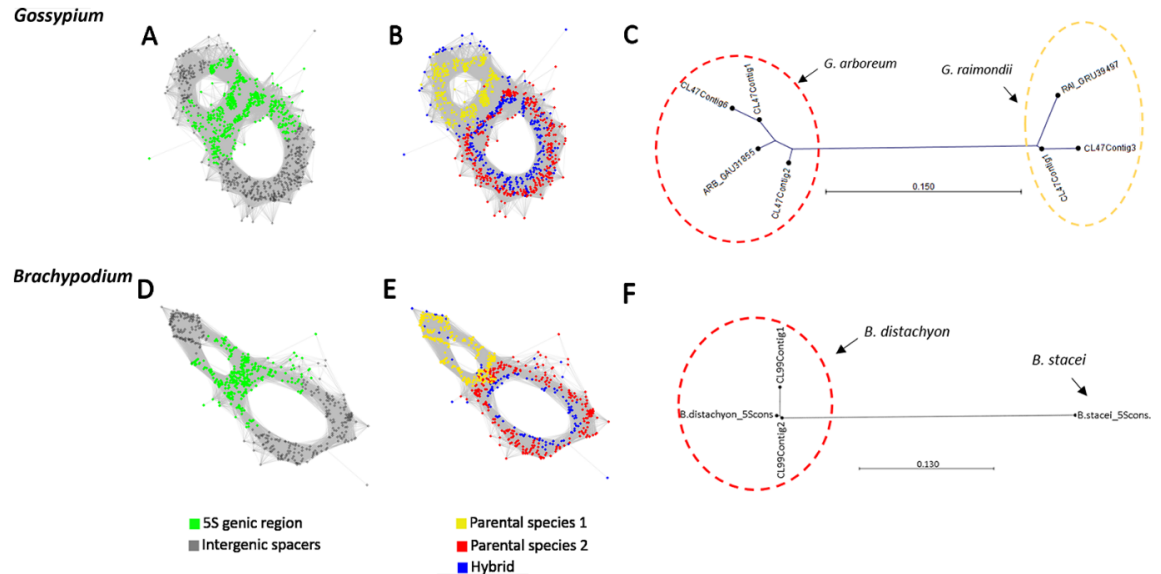
- single species
- comparative analysis

Repeat clusters as a reference

- ChIP-seq
- assembly
- graph shapes

The Utility of Graph Clustering of 5S Ribosomal DNA Homoeologs in Plant Allopolyploids, Homoploid Hybrids, and Cryptic Introgressants

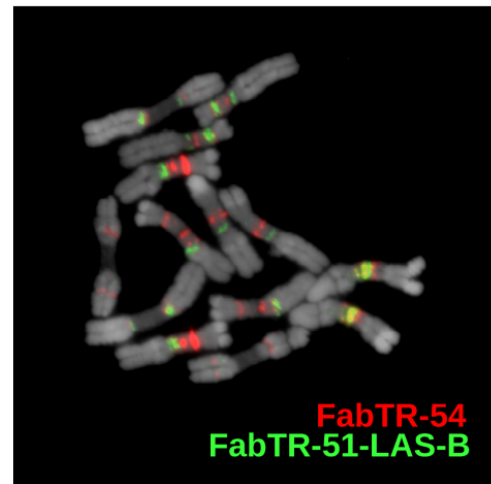
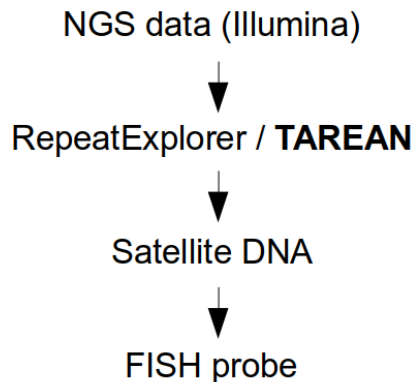
Sònia García^{1,2}, Jonathan F. Wendel³, Natalia Borowska-Zuchowska⁴, Malika Ainouche⁵, Alena Kuderova² and Ales Kovarik^{2*}



Applications

- Repeat composition
 - single species
 - comparative analysis
- Repeat clusters as a reference
 - ChIP-seq
 - assembly
 - graph shapes
- Satellite DNA
 - cytogenetic studies

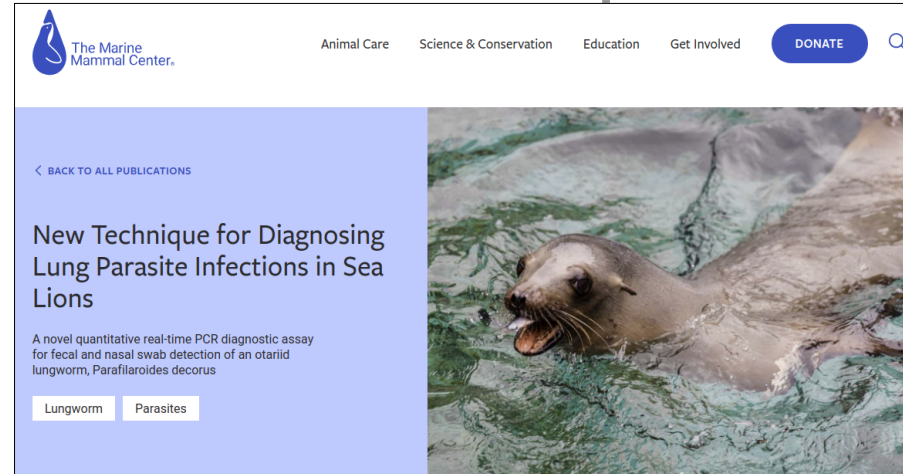
Cytogenetic markers (FISH probes)



Lathyrus sativus
(FISH by L. Avila Robledillo)

Applications

- Repeat composition
 - single species
 - comparative analysis
- Repeat clusters as a reference
 - ChIP-seq
 - assembly
 - graph shapes
- Satellite DNA
 - cytogenetic studies
 - diagnostic markers



Applications

- Repeat composition
 - single species
 - comparative analysis
- Repeat clusters as a reference
 - ChIP-seq
 - assembly
 - graph shapes
- Satellite DNA
 - cytogenetic studies
 - diagnostic markers
- RE utilization in other pipelines

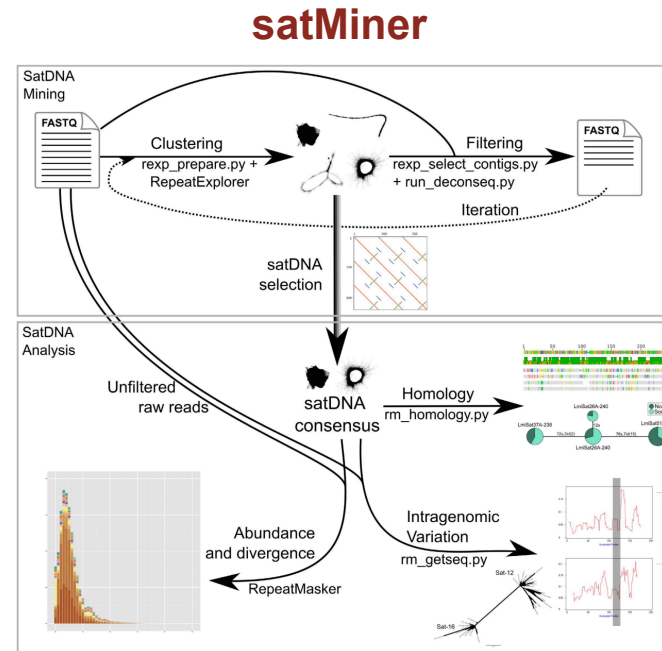
SCIENTIFIC REPORTS

OPEN

High-throughput analysis of the satellitome illuminates satellite DNA evolution

Received: 14 January 2016
Accepted: 02 June 2016

Francisco J. Ruiz-Ruano, María Dolores López-León, Josefa Cabrero & Juan Pedro M. Camacho



Applications

Repeat composition

- single species
- comparative analysis

Repeat clusters as a reference

- ChIP-seq
- assembly
- graph shapes

Satellite DNA

- cytogenetic studies
- diagnostic markers

RE utilization in other pipelines

Mann et al. *BMC Bioinformatics* (2022) 23:40
<https://doi.org/10.1186/s12859-021-04545-2>

BMC Bioinformatics

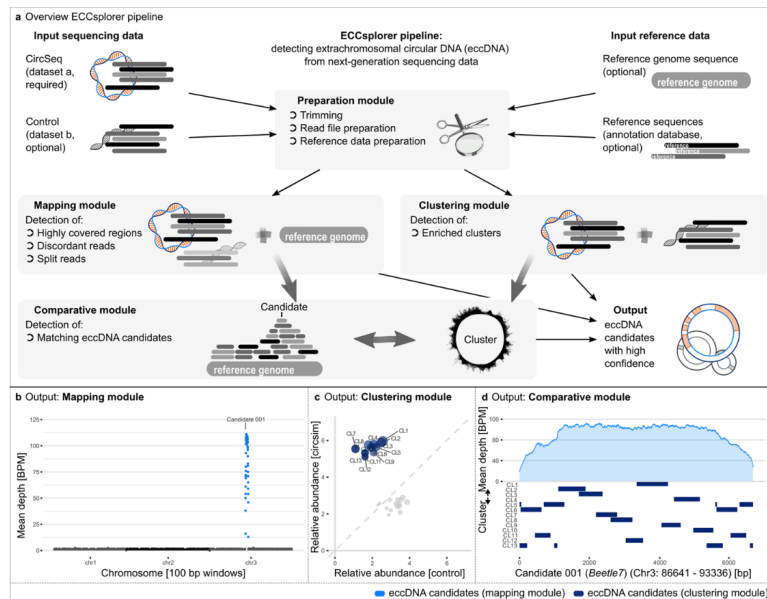
SOFTWARE

Open Access



ECCsplorer: a pipeline to detect extrachromosomal circular DNA (eccDNA) from next-generation sequencing data

Ludwig Mann[✉], Kathrin M. Seibt[✉], Beatrice Weber[✉] and Tony Heitkam[✉]



Applications

- Repeat composition
 - single species ***
 - comparative analysis ***
- Repeat clusters as a reference
 - ChIP-seq ***
 - assembly ***
 - graph shapes*
- Satellite DNA
 - cytogenetic studies ***
 - diagnostic markers
- RE utilization in other pipelines



Enjoy the workshop !